

# **ServiGo Application**

## **Project Documentation**

## **i. Acknowledgement**

I would like to express my gratitude to all those who supported me to build this ServiGo app successfully and who guided me throughout development. I would like to specially thanks to my lecturers for their greatest guidance and knowledge. Also my friends who supported me and my family for their encouragement. Their support made it possible to complete ServiGo app successfully.

## **ii. Abstract**

ServiGo is a mobile app that developed to connect home service seekers and skilled technicians. This system is provide a solution as end to end with search services, booking service, secure online payments and real time tracking.

This project included two platforms. An android mobile app for customers and technicians and a website for administrators. The android mobile app is developed using java and fragment architecture. Backend is developed using PHP and RESTful APIs with MySQL. This project uses firebase authentication, firebase firestore, firebase storage for store user profile images, real time data updating and push notifications

ServiGo represent a modern mobile application and website that uses modern technologies to ensure reliable, efficient and user friendly experience to users.

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# 1. Introduction

## 1.1. Project Overview

ServiGo is a modern service booking platform (M-Commerce) that developed to connect home service seekers and well skilled technicians. It is a very efficient and reliable platform. ServiGo project is a dual interface system.

1. Android mobile application - This application for the customers to book services and for technicians to manage assignments.
2. Laravel web admin panel - This application is for admin to handle bookings, oversee the system and manage users.

The goal of the ServiGo system is to simplify day to day tasks by providing end to end experience for users from browsing services, scheduling appointments, making secure payments and real time location tracking on google map.

## 1.2. Objectives

- To simplify service selection and booking services through user friendly modern platform.

ServiGo is developed to find services without any confusion. The interface of ServiGo is very simple and user friendly, it allows customers to choose the service easily and schedule appointment in just few steps.

- To provide a seamless and real time booking experience.

The ServiGo allows users to book services smoothly without any delays and without confusions. With real time updates, user able to see available options instantly. That ensure efficiency and reliability of booking process.

- To enable real time technician tracking.

ServiGo allows customers to track technician live location smoothly on google map after technician assigned. This ensure guarantee of the system and it helps customer to know exactly when technician will arrive.

- To provide secure, non confusion and reliable cashless payment.

The ServiGo platform implements a secure and trusted payment gateway to payment process. This allows the customers to make checkout experience safely and smoothly without cash and any issue. Also ensures transparency and reduces risks with physical transaction.

- To enhance customer trust about the ServiGo services through the rating and reviews of the services.

In ServiGo app, customer able to see others reviews and ratings for the services and also they can leave their feedback after service is completed. This ensure the service quality and build the trust in platform.

- To Control full system by administrator.

The admin panel of ServiGo is allows administrators to manage user accounts and manage bookings, assign technicians, monitor system and handle failures. This ensures the ServiGo system is runs smoothly and efficiently without any issues and it helps to identify issues.

- Collect data to analyze and get decisions.

The ServiGo system collects data and organize the data related to bookings, users and services. This helps administrators and developers to improve the ServiGo system in future.

## **2. Background**

In present, world is fast and people growingly rely on digital platforms to simplify their day to day tasks. But finding a professional skilled, reliable providers for the tasks such as painting, plumbing, carpentering, cleaning are very difficult now. Traditionally searching service providers by personal recommendations or by advertisement. It is leads to time wasting, unreliable and not guaranteed.

Many service platforms are providing simple solutions but not the complete solution. Some platforms runs without any verification of technicians, wasting users time, do not provide any guarantee. It may leads to fail the whole system. Some users face to difficulties with scheduling services, tracking technician location, trust, quality and efficiency.

In today's fast growing of the mobile technology and cloud based services gives opportunity to improve the process of how the service providing platforms are runs. The real time database, real time tracking and mobile payments can implement to service providing platforms to make more efficient, reliable and more user friendly experience to users.

ServiGo is developed to directly deal with these challenges by providing trust full, reliable guaranteed platform that connects customers and skilled technicians. ServiGo developed by using modern technologies such as cloud services, real time tracking to provide smooth experience. The features of ServiGo like review system, live technician tracking and secure payments without physical cash leads to improve efficiency, reliability, transparency and user friendly experience.

## **3. Specifications**

### **3.1. User Roles**

#### **Customer**

- Customer able to sign in and sign up.
- Search services from home page.
- View booking history.
- Customer can refresh the booking history by shaking the phone.
- Add profile image, update thier profile details.
- Booking a service by set date, time, address, mobile, email address and name

- Make secure payments with payhere.
- Track technician location in real time.
- Customer can call to the assigned technician.
- Leave review and submit rating after service is completed.
- Customer able to receive emails and notifications.
- Customer able to log out.

### **Technician**

- Customer able to sign in and sign up.
- Add profile image, update profile details.
- View assigned bookings.
- Technician can refresh the booking history by shaking the phone.
- Technician can call to the customer
- View booking history.
- View customer location on google map.
- Change the status of the booking after arrived and completed.
- Share real time location with customers.
- Technician able to Log out from ServiGo.

### **Admin**

- Log in to the admin panel.
- Manage users (ban / unban).
- Oversees users, bookings.
- Update status of bookings.
- Assign technicians to the bookings
- Monitor system and analyze.
- Delete bookings.

## **3.2. Features and Functionalities**

### **Mobile Application**

- **Authentication**

The customers and technicians are allowed to log in to the ServiGo by using firebase authentication.

- **Find services**

Customer allows to find services from search bar, services category fragment and home fragment. Customer able to view details by clicking a services and in that space show ratings and reviews also.

- **Booking System**

Customers can booking a service by selecting date,time and address. After placing the booking, the address is automatically converting to latitude and longitude by Geocoding API. Users can refresh booking history by shaking the phone.

- **Payments**

Payments are processing by using payhere with LKR.

- **Real Time Tracking**

Customer allows to track the assigned technician location through the Google map after technician assignment. Also technician can view customer location.

- **Rating and Reviews**

Customer can rate and leave a review after complete the service. It will be displaying on the detailed service fragment.

- **Notifications**

Customer and technician allows to get notification after sign in and sign up from firebase cloud messaging. Customer receives a email after technician assignment by admin.

- **Assigned bookings**

Technician can view assigned bookings on technician dashboard. Technician able to change the status of booking such as arrived and completed.

### **Admin**

- **Authentication**

Only administrators allows to login to the admin panel.

- **User Management**

Admin allows to view user details, can ban unban.

## ● **Booking Management**

Administrator can assign technician to each booking, allows to change the status of the booking and can delete booking.

### **3.3. Functional Requirement**

The functional requirements represent main operations of ServiGo system.

- The ServiGo system allows users to sign in and sign up using Firebase Authentication.
- The ServiGo system allows customers to search services.
- The ServiGo system allows customers to place a booking by choosing date, time and address.
- System shall allow to convert user address to geocode using Geocoding API.
- System shall allow customers to make payment securely with PayHere.
- ServiGo system shall store booking details only after the customer confirmation payment is successful.
- System allows users to update their profile details.
- The ServiGo shall allow technicians to view assigned bookings.
- System allows technician to change booking status (arrived, completed).
- System shall allow to track real time location of technician on google map.
- System shall allow customers to leave feedback about the service after service is completed by technician.
- System shall allow admin to login to the admin panel.
- System shall allow admin to manage user status as ban and unban.
- System shall allow admin to assign technician to each booking.
- System shall allow admin to change booking status.
- System shall allow admin to delete booking.

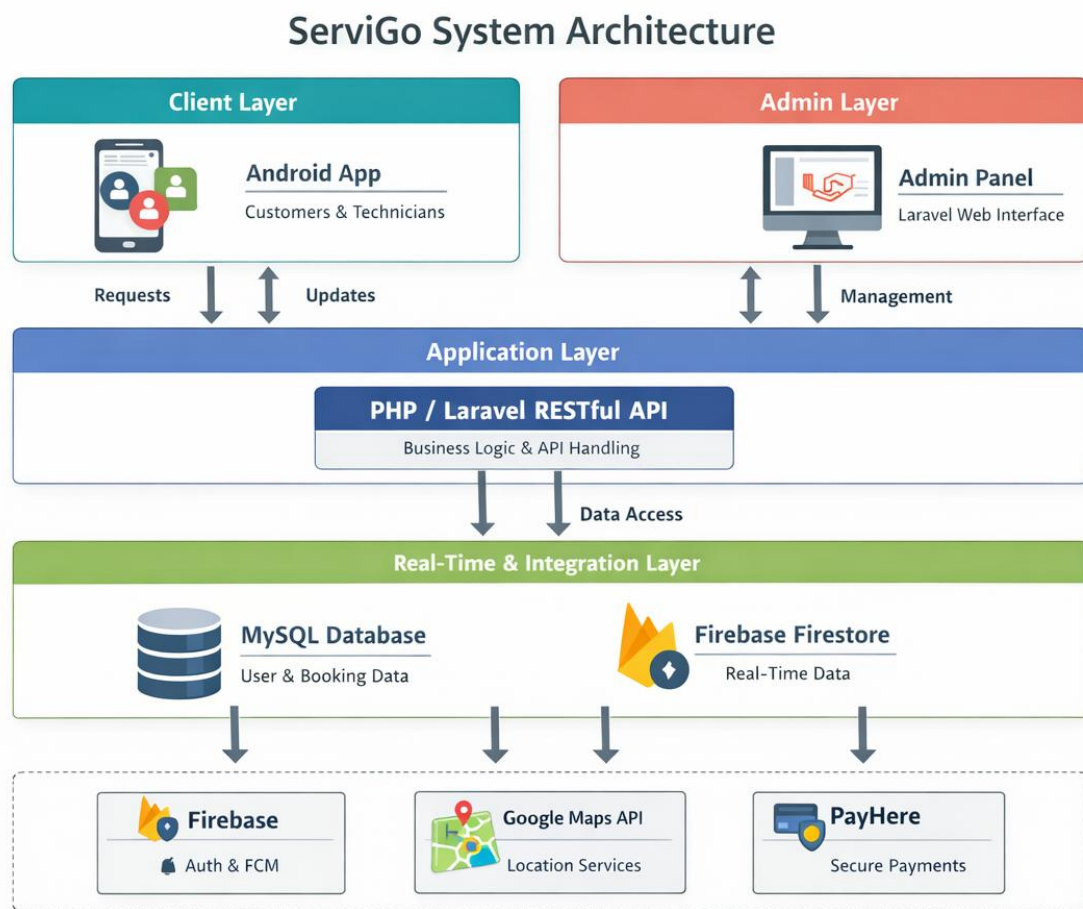
### **3.4. Non functional Requirements**

- Performance : ServiGo system should respond to users actions within limited time period.
- Scalability: ServiGo should able to manage large amount of booking and users.
- Security: User personal details and transactions must handle securely and efficiently by using secure APIs and authentication.
- Reliability: ServiGo should run and operate continuously without any issue.

- Usability: ServiGo application should provide user friendly, modern and simple interface.
- Availability: The ServiGo application should be accessible in anytime.
- Maintainability: ServiGo system should be easy to update and maintain.
- Compatibility: ServiGo application shall be run in devices that API level is 28 and above.

## 4. Design

### 4.1. System Architecture



**Figure 1: System Architecture**

#### 1. Client Layer

This includes mobile application used by customer and technician. In this layer customer can brows service, book service, make payments and track locations. Technician can view

customer location, change the status of booking. This layer sends requests to the backend for the specific operations.

## 2. Admin Layer

This layer represent by laravel application (Admin panel) that use MVC architecture. Admin can ban and unban users, assign technicians to bookings, update booking statuses, delete bookings and oversees the system from dashboard.

## 3. Application Layer (Backend API)

The backend of ServiGo is developed by using PHP and RESTfull architecture. It is handling bussines logics, handle client requests, make communication between mobile application, mysql database, firebase firestore and admin panel.

## 4. Data and Real Time Layer

This layer contained all data of the system. MySQL database stored such as admins, bookings, reviews. Firebase firestore stored users, technician locations, services. Firebase storage is used to store banner pictures and profile picture of the users.

## 5. Integration Layer

This layer include third party services such as firebase, google map APIs, PayHere.

## 4.2. Technology Stack

| Layer              | Technology                                                                                                        |
|--------------------|-------------------------------------------------------------------------------------------------------------------|
| Mobile App         | Java (Android), XML Layouts, Material Design3, ViewBinding, Advanced Components, Telephony, Sensors (Motion, GPS) |
| Backend            | PHP REST API, MySQL                                                                                               |
| Admin Panel        | Laravel Framework(11, PHP 8.5v), Blade Templates, Eloquent ORM, Tailwind CSS                                      |
| Real Time Services | Firebase Authentication, Firestore, Cloud Messaging, Storage                                                      |
| Maps & Location    | Google Maps SDK, Geocoding API, Play Services Location                                                            |
| Payments           | PayHere SDK                                                                                                       |
| Media Handling     | Glide, Picasso                                                                                                    |
| Networking         | Retrofit 2 (Gson and Scalars)                                                                                     |

**Table 1: Technology Stack**

### 4.3. Project Architecture

| Package                     | Description                                                             |
|-----------------------------|-------------------------------------------------------------------------|
| lk.teveram.servigo.activity | Main Activity entry points (Splash, Login, Maps, Dashboards).           |
| lk.teveram.servigo.fragment | UI components (Home, Booking Form, Profile, Service Details).           |
| lk.teveram.servigo.api      | API client configurations and endpoint definitions.                     |
| lk.teveram.servigo.model    | POJO classes for data mapping (User, Booking, Review, ServiceCategory). |
| lk.teveram.servigo.adapter  | RecyclerView adapters (Bookings, Services, Reviews).                    |

**Table 2: Project Architecture**



## 4.4. Use Case Diagram



Figure 2: Use Case Diagram

## 4.5. Class Diagram

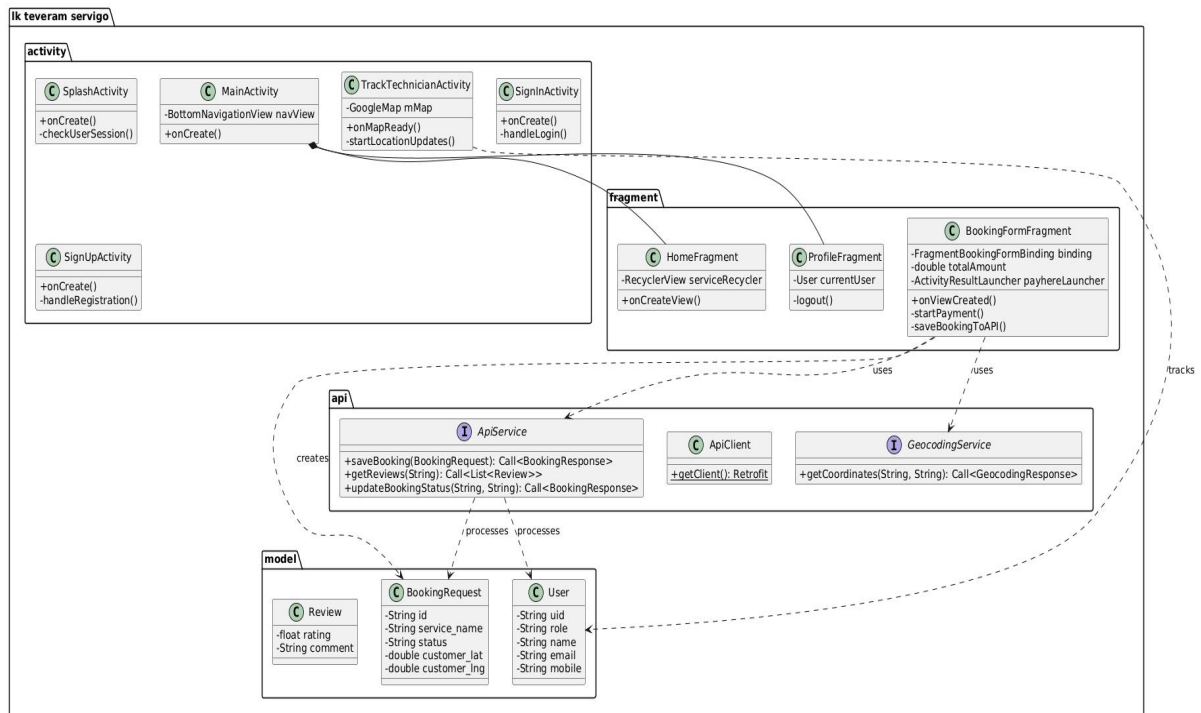


Figure 3: Class Diagram

#### 4.6. Sequence Diagram

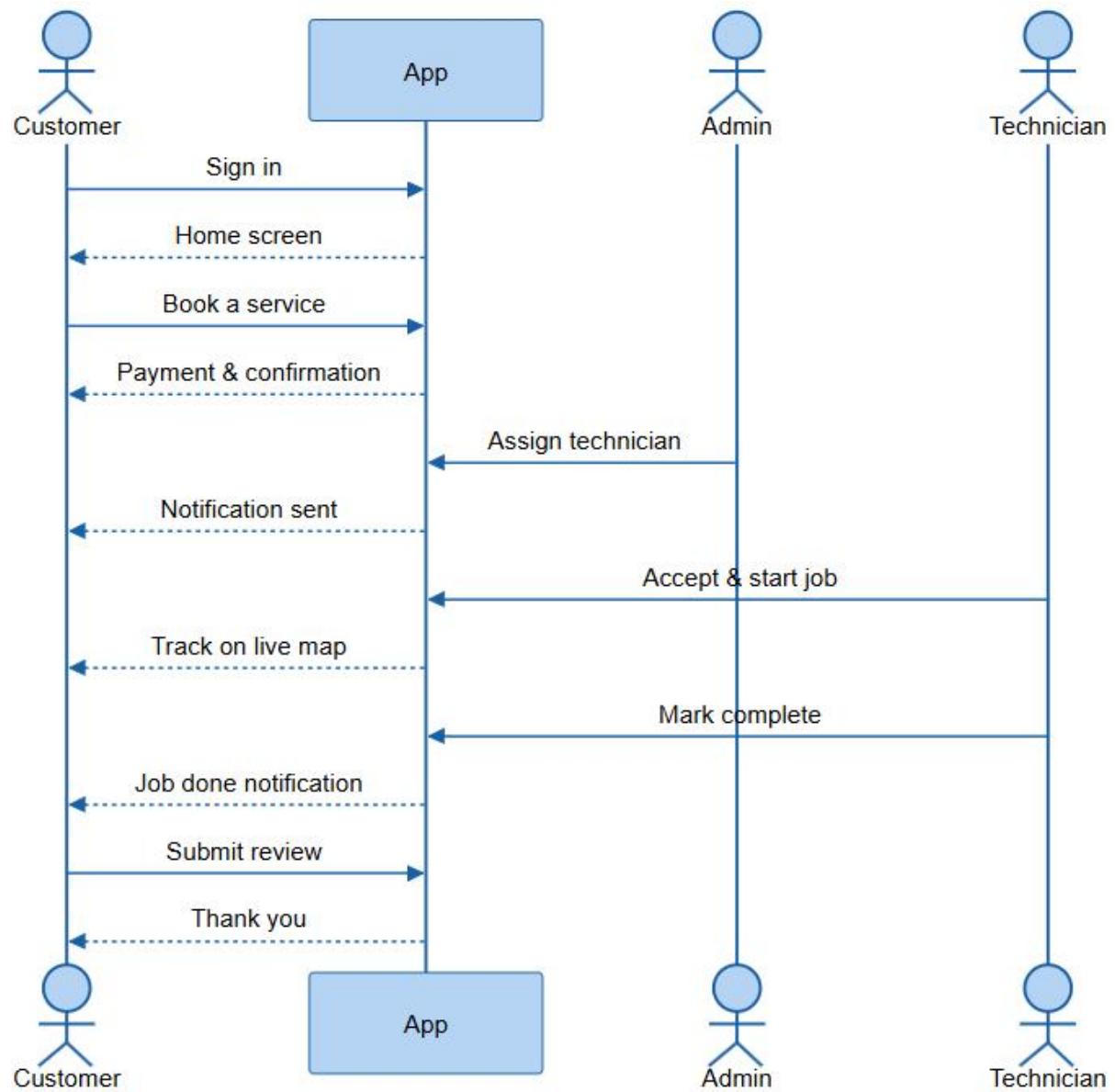


Figure 4: Sequence Diagram

#### 4.7. ER Diagram (MySQL)

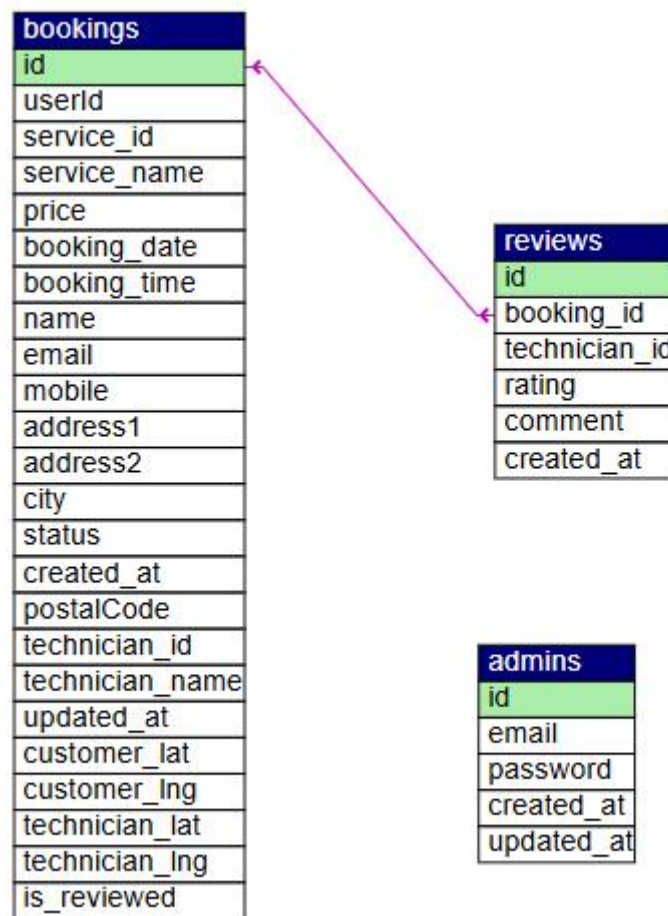


Figure 5: MySQL ER Diagram

#### 4.8. ER Diagram (Firebase Firestore)

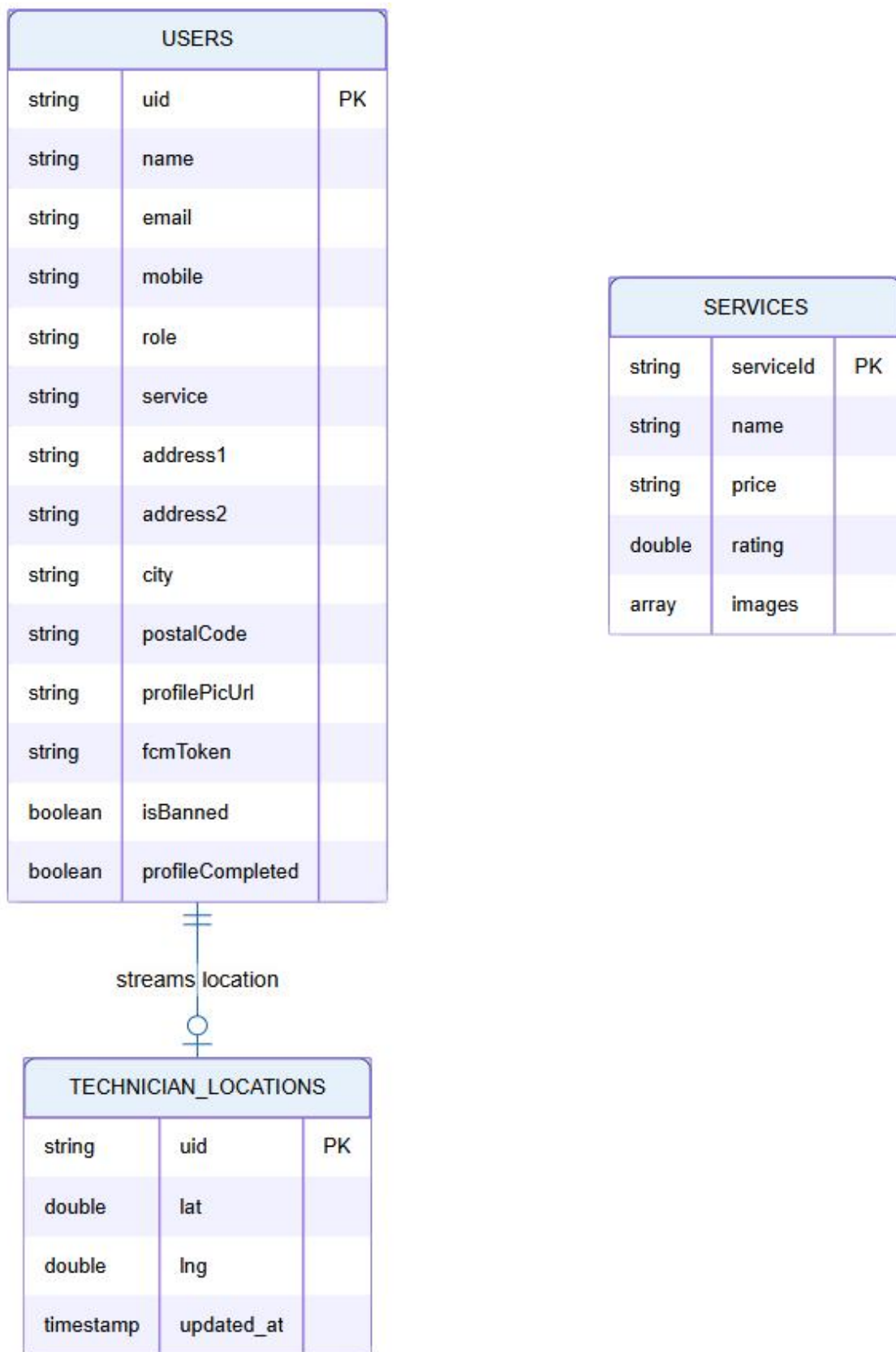


Figure 6:ER Diagram (Firebase)

## 4.9. Combined ER Diagram (MySQL + Firebase Firestore)

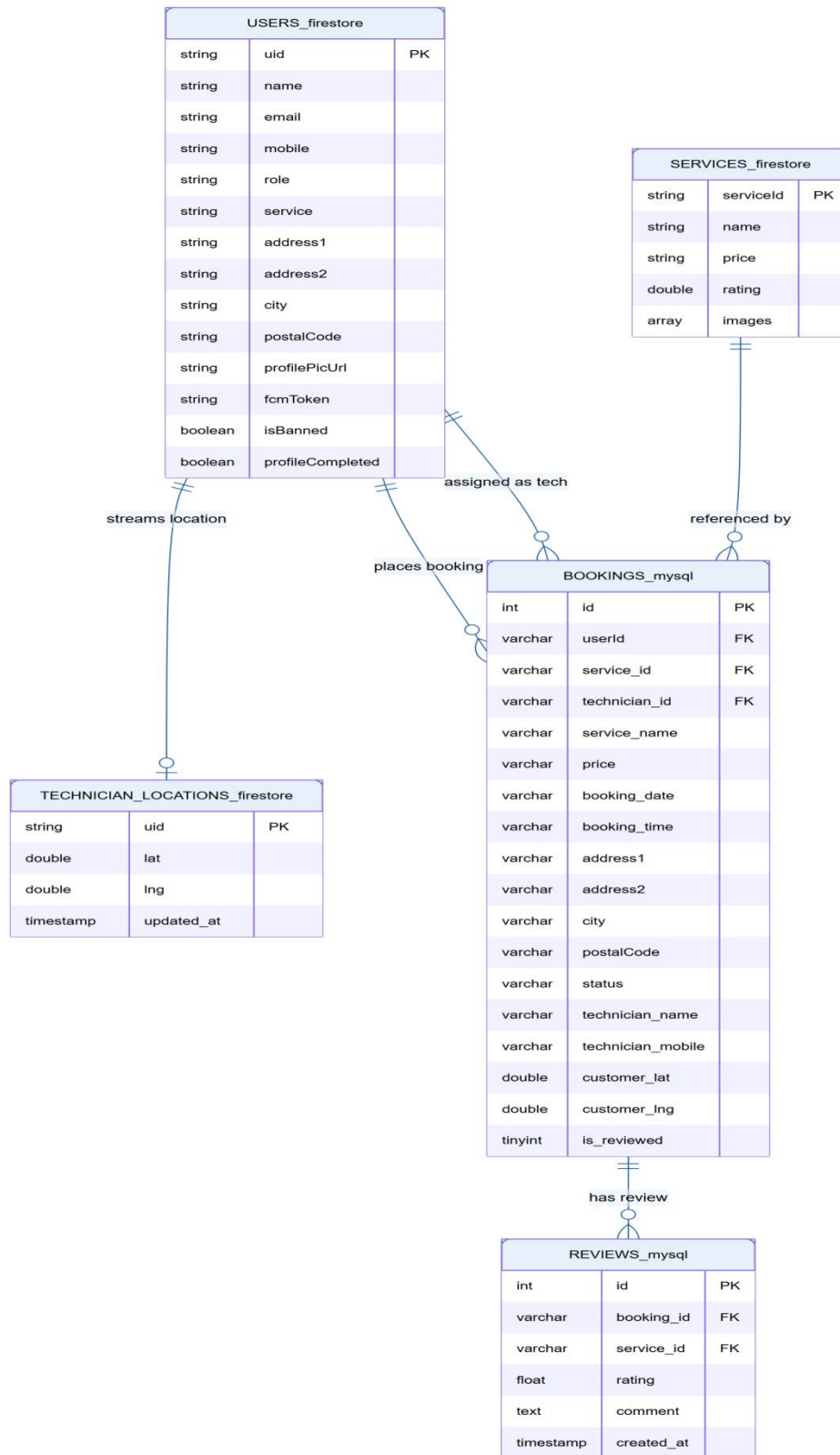


Figure 7: Combined ER Diagram (MySQL + Firebase)

## 4.10. User Interface (UI) Design

### Splash Screen

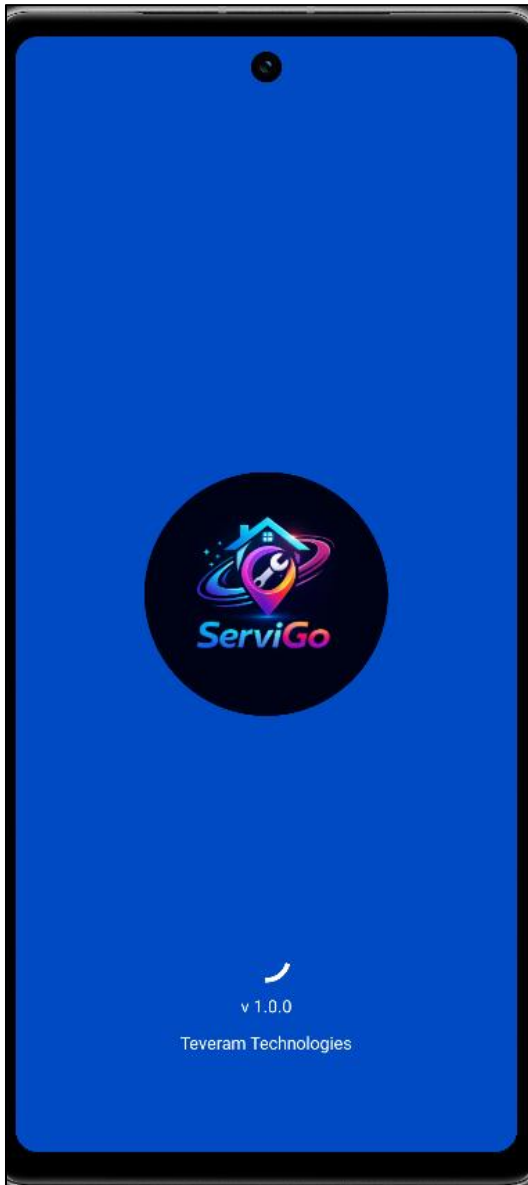


Figure 8 : Splash Scree Light Theme



Figure 9 : Splash Screen Dark Theme

This is the Splash Screen of the Servigo. The Application starts with this screen.

## Sign In Screen

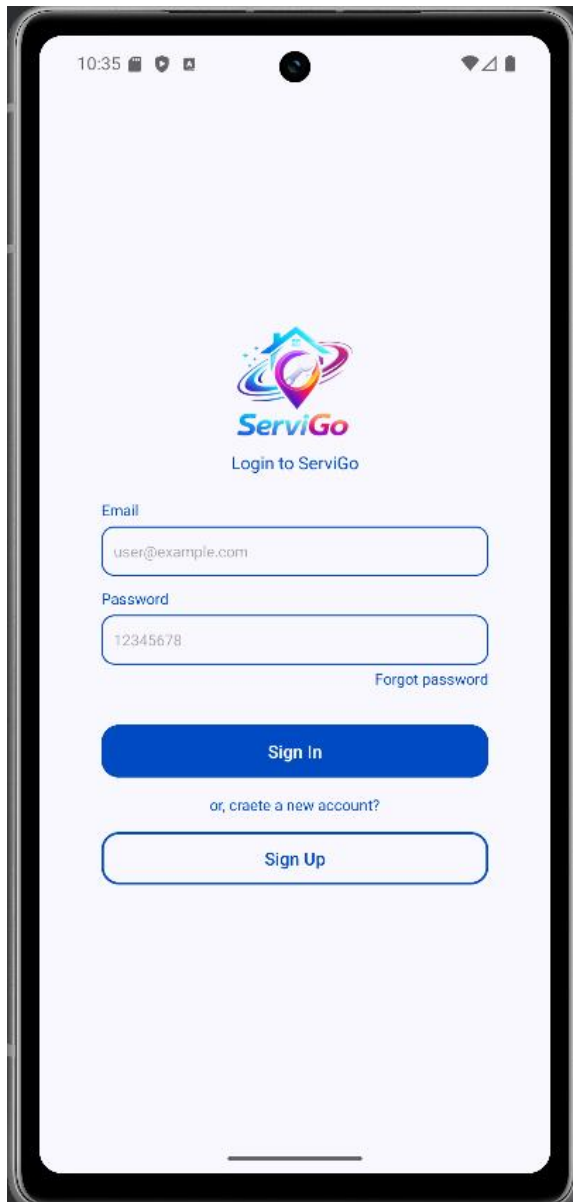


Figure 10: Sign in Light theme

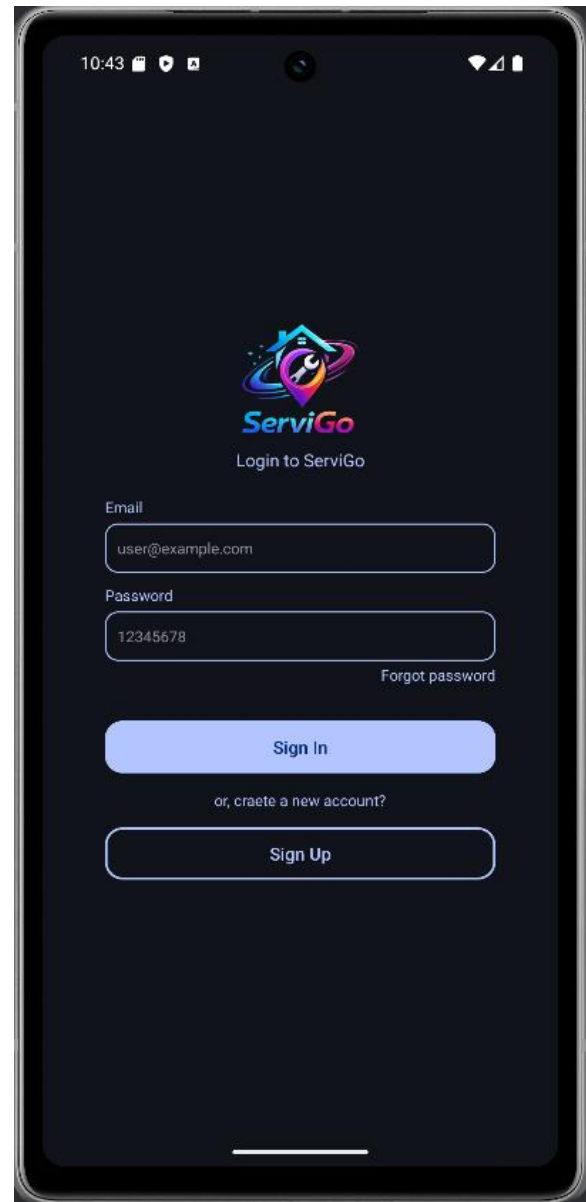
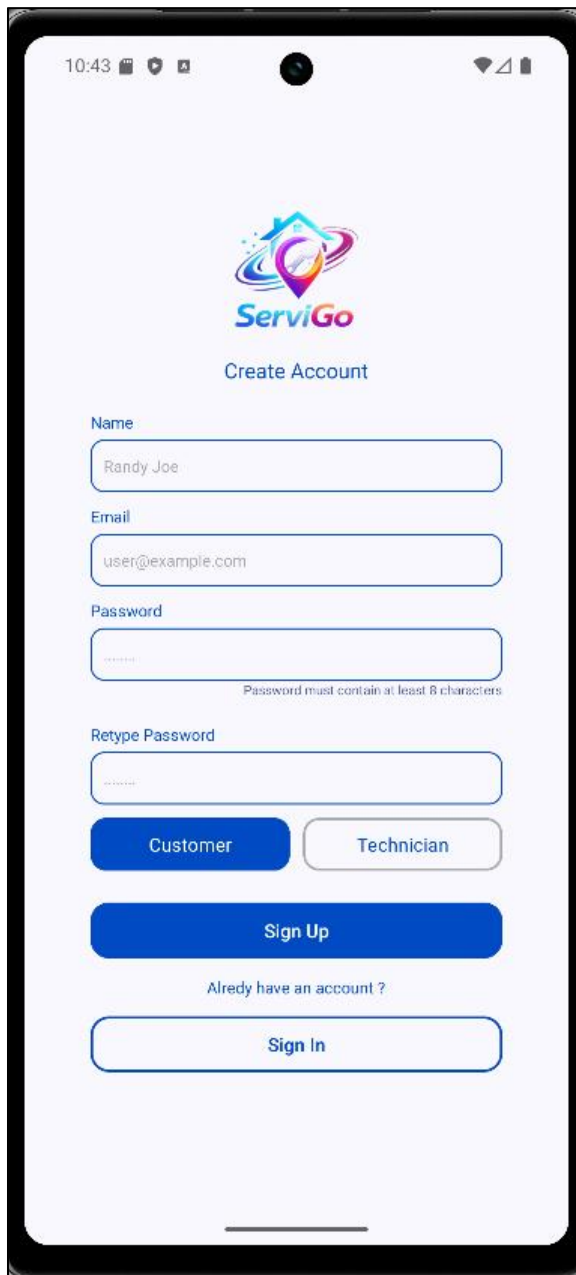


Figure 11: Sign In Dark Theme

This is the Sign screen of the ServiGo. After the splash screen user can see this screen and if user already registered, user can sign in from here. If user is new to ServiGo user can touch Sign Up button and it redirects to sign up screen.



## Sign Up Screen



10:43

**ServiGo**

Create Account

Name

Randy Joe

Email

user@example.com

Password

.....

Password must contain at least 8 characters

Retype Password

.....

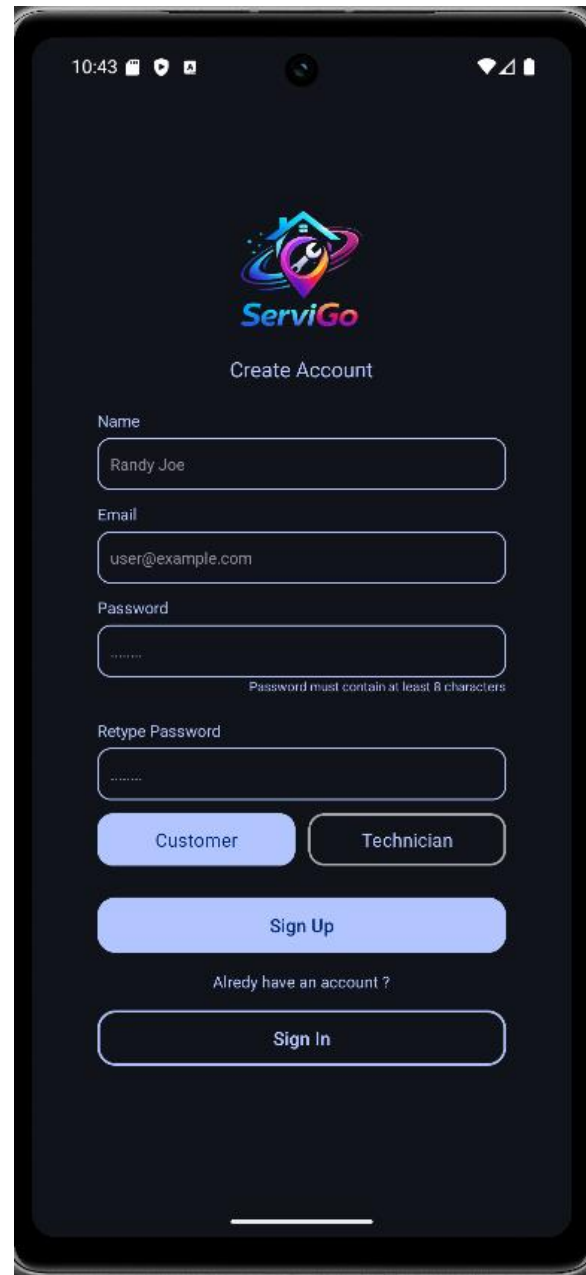
Customer Technician

Sign Up

Alredy have an account ?

Sign In

Figure 12 : Sign Up Screen Light



10:43

**ServiGo**

Create Account

Name

Randy Joe

Email

user@example.com

Password

.....

Password must contain at least 8 characters

Retype Password

.....

Customer Technician

Sign Up

Alredy have an account ?

Sign In

Figure 13 : Sign Up Screen Dark

This is the Sign Up Screen of the ServiGo App. After clicking Sign up button on Sign In screen, user will see this screen. In this screen user can select the role and must enter the details to create new account. All the input fields are validated.

## Customer Home Screen

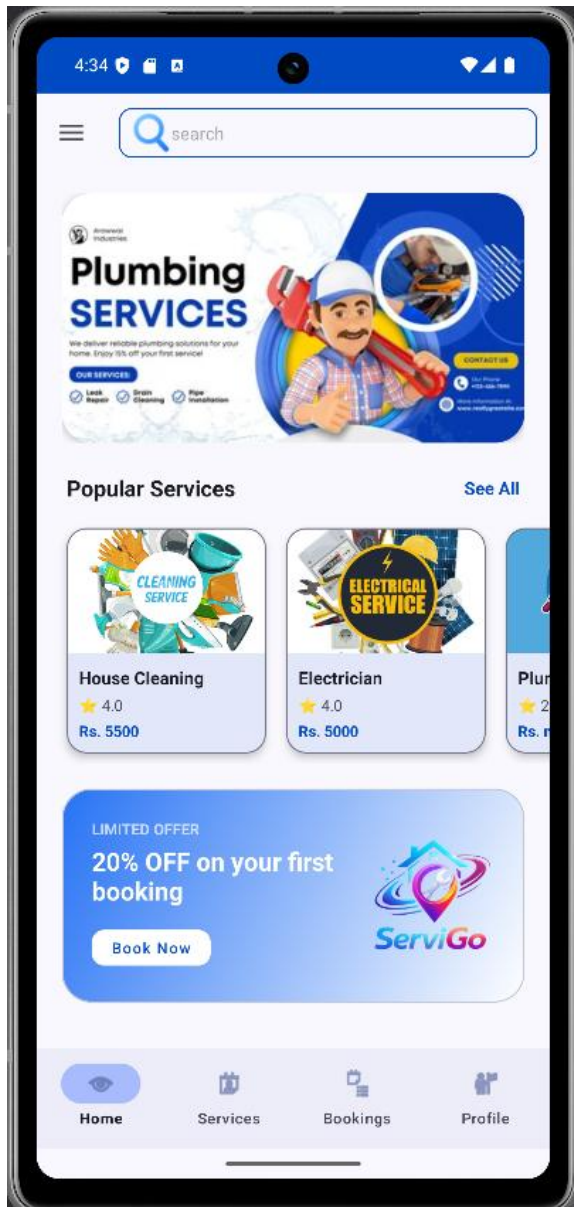


Figure 14: Customer Home screen Light theme

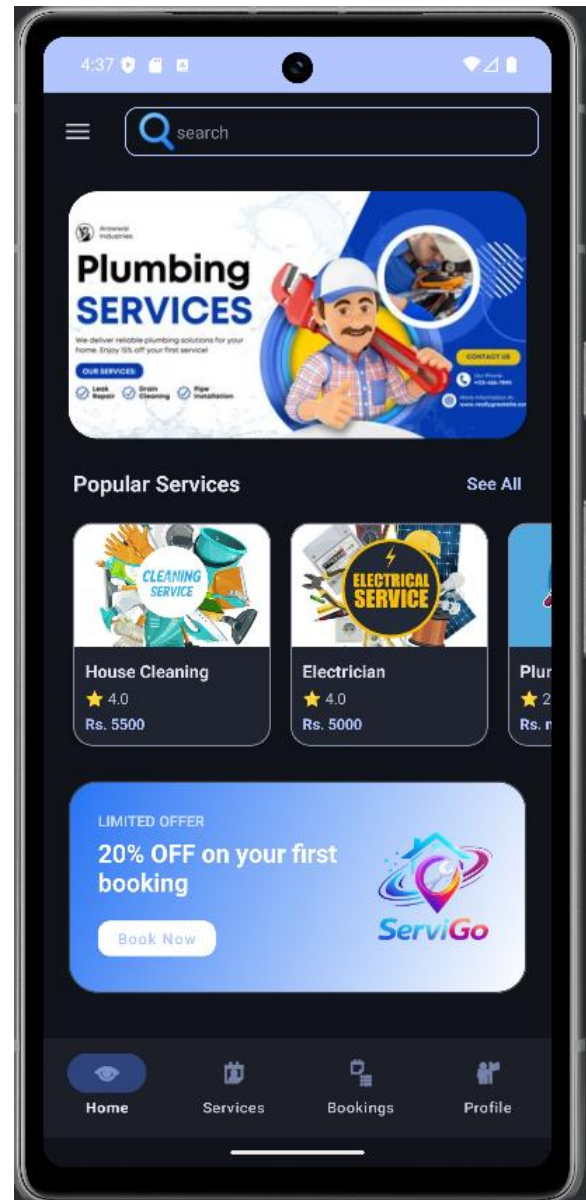


Figure 15: Customer Home screen Dark theme

After customer sign in, customer can view the home screen of ServiGo application. In this screen, customer can see automated image slider, popular services based on customer ratings, banner. With scrolling down customer can see all services section. In the header there is a search bar the allows user to search any services. After search services it will show in this screen. The hamburger icon of the menu leads to open side navigation bar. Customer allows to navigate between screens by using bottom navigation bar.

## Side Navigation Bar

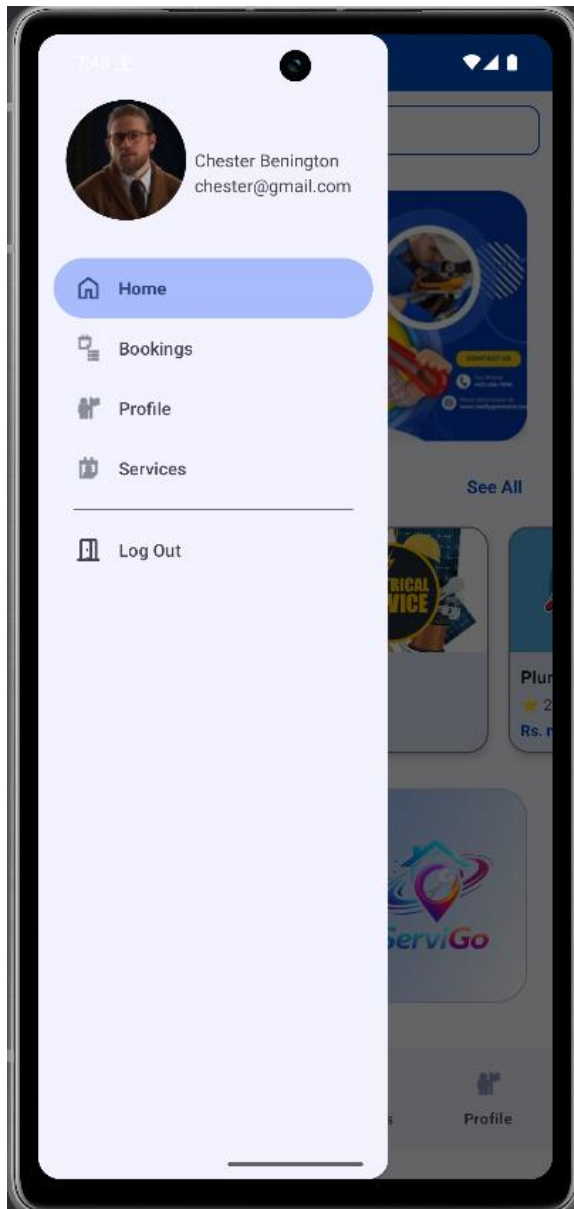


Figure 16: Side navigation bar Light Theme

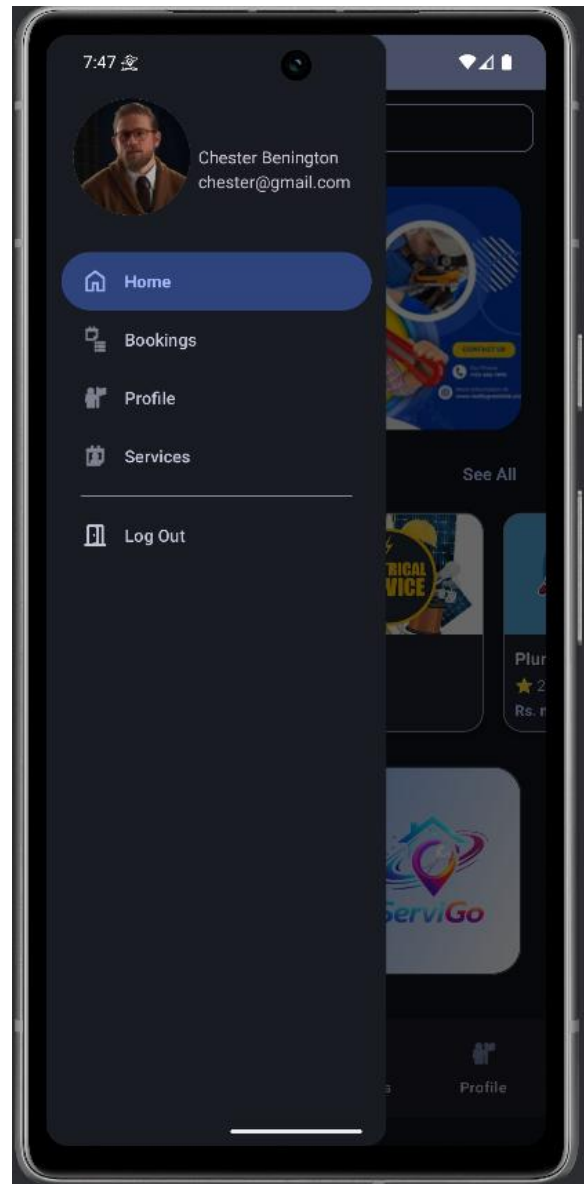


Figure 17: Side navigation bar DarkTheme

This is the side nav bar of ServiGo. It shows navigation buttons to Bookings, profile, Services. Also user can logout by logout button. Top of the side bar shows user full name, email and the profile picture.

## Services Screen

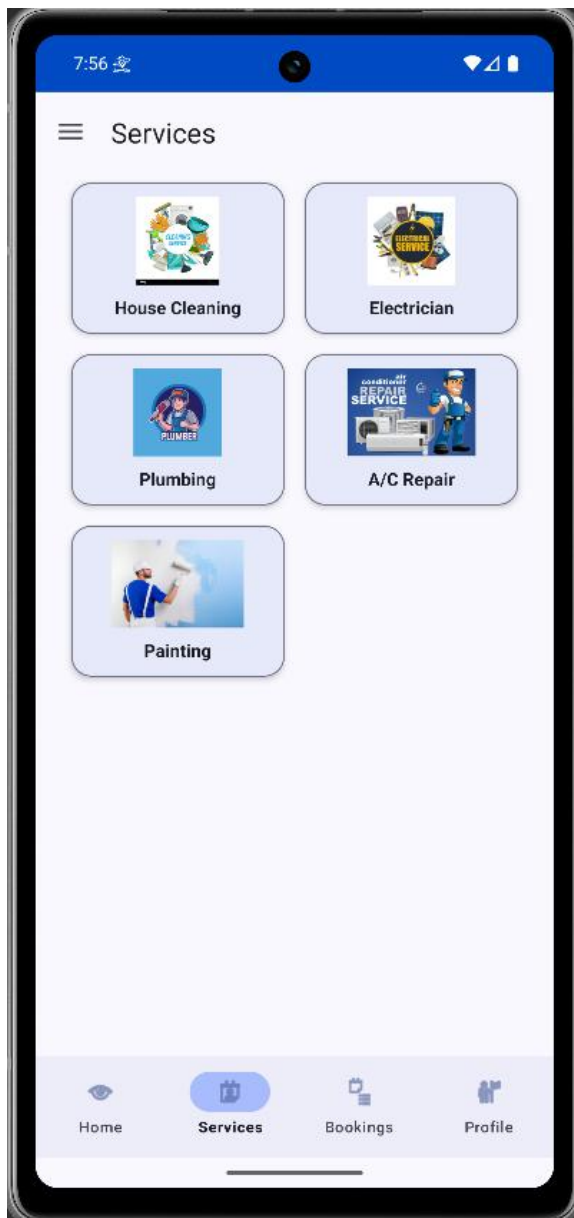


Figure 18: Services Screen Light Theme

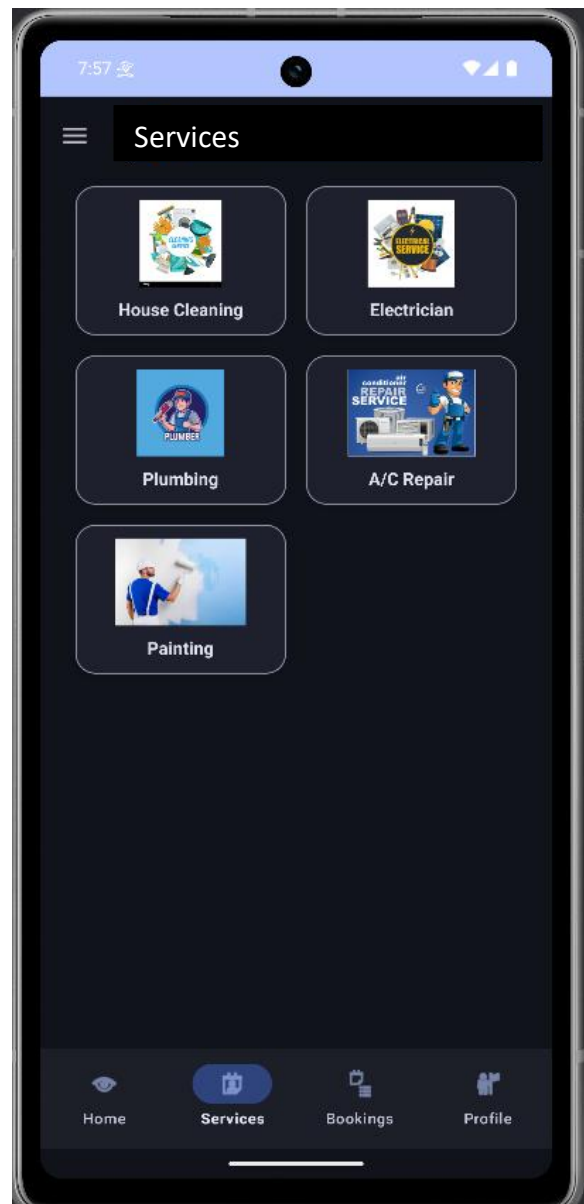


Figure 19: Services Screen Dark Theme

In this screen customer able to see all the services and allows to view details of each services by touching s service.

## Single Service Screen

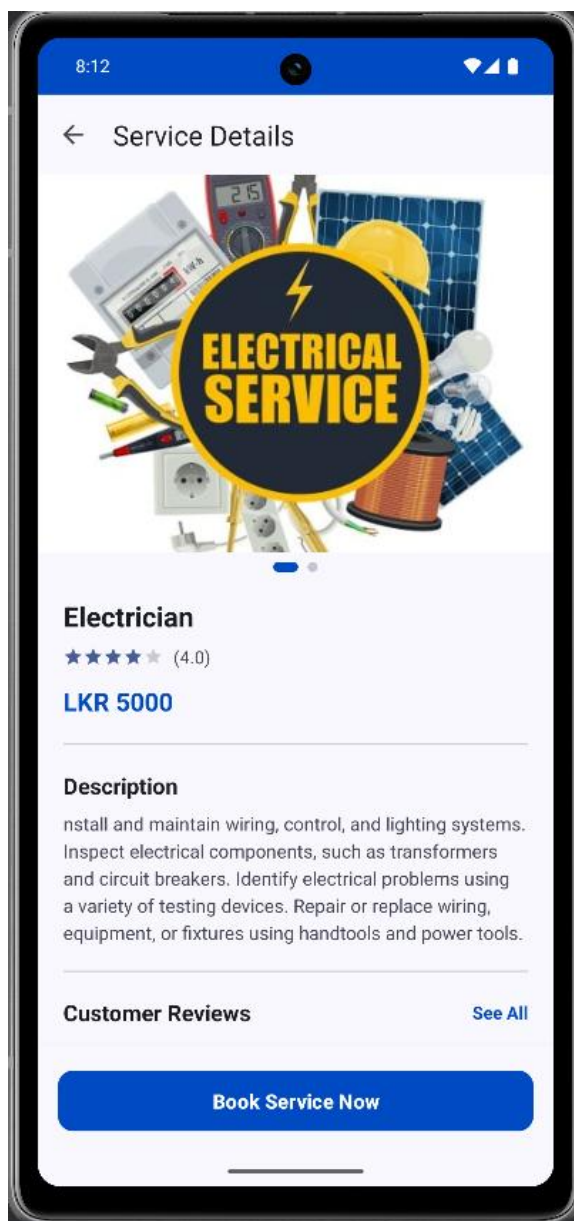


Figure 20: Single service screen light theme

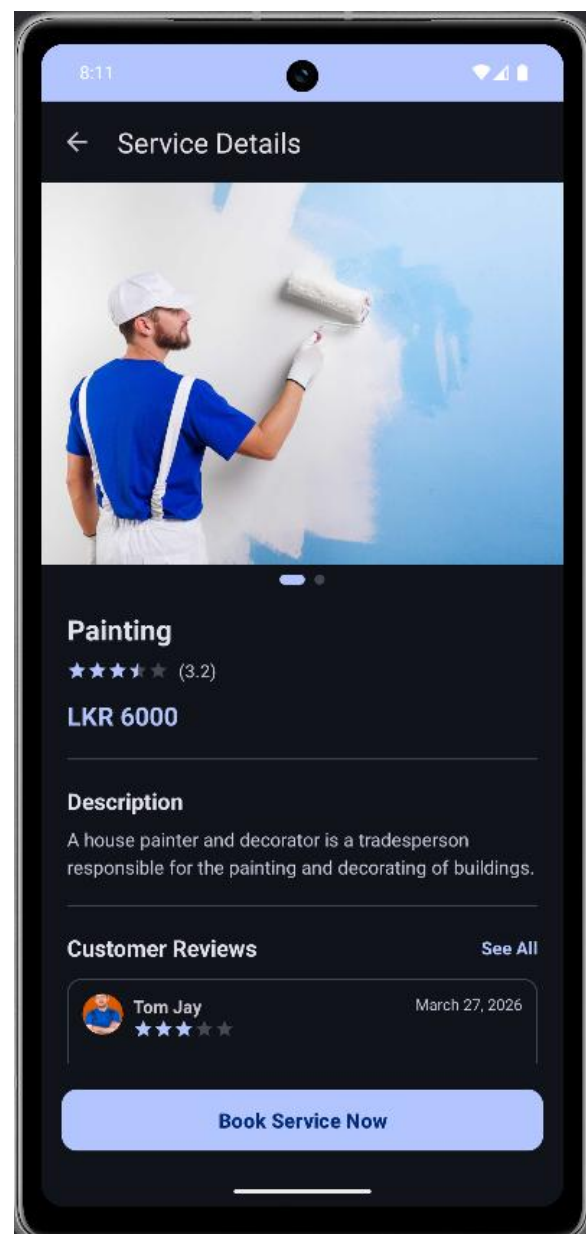


Figure 21: Single service screen light theme

This screen shows details of the service, reviews of customers and ratings. User can fill the booking form by touching book service now button.

## Booking Form screen

The image displays two mobile application screens for a service booking form. Both screens have a blue header with the time 8:18 and status icons. A search bar is located at the top of each screen.

**Left Screen: Service Booking**

- Contact Information:**
  - Full Name: Chester Benington
  - Email Address: chester@gmail.com
  - Contact Number: 0776666963
- Appointment Schedule:**
  - Date: 31/3/2026
  - Time: 14:30
- Service Location:**
  - Custom Address toggle: Off
  - Address Line 1: (empty)
- Pay and Confirm Booking** button
- Bottom Navigation:** Home, Services (selected), Bookings, Profile

**Right Screen: Service Booking**

- Service Location:**
  - Custom Address toggle: On
  - Address Line 1: No.1, Bandaranayaka Pura
  - Address Line 2: Rajagiriya
  - City: Colombo
  - Post Code: 10100
- Order Summary:**
  - Selected Service: Electrician
  - Appointment: 31/3/2026 at 14:30
  - Total Amount: LKR 5000
- Pay and Confirm Booking** button
- Bottom Navigation:** Home, Services (selected), Bookings, Profile

Figure 22:Booking Form

This is the booking form. The bookings are storing based on this details. Customer must fill this form to confirm the booking. Customer can add new address to the booking by toggling the custom address toggle.



## Payment Phase

The image displays two smartphone screens side-by-side, illustrating the 'Payment Phase' of a 'Service Booking' application. Both screens have a blue header bar with the time (8:24 on the left, 8:26 on the right) and standard mobile status icons. A search bar is located at the top of each screen.

**Left Screen (8:24):**

- Service Booking** header.
- Contact Information** section with fields for 'Full Name' (filled with 'Chester Benington') and 'Email Address' (filled with 'chester@gmail.com').
- Pay with** section, featuring a 'Secured by PayHere' badge and a 'Sandbox' label.
- Bank Card** section with logos for VISA, Mastercard, American Express, DISCOVER, and Diners Club International.
- Other** section with logos for genie, eZcash, m-Cash, Frappé, and a QR code.

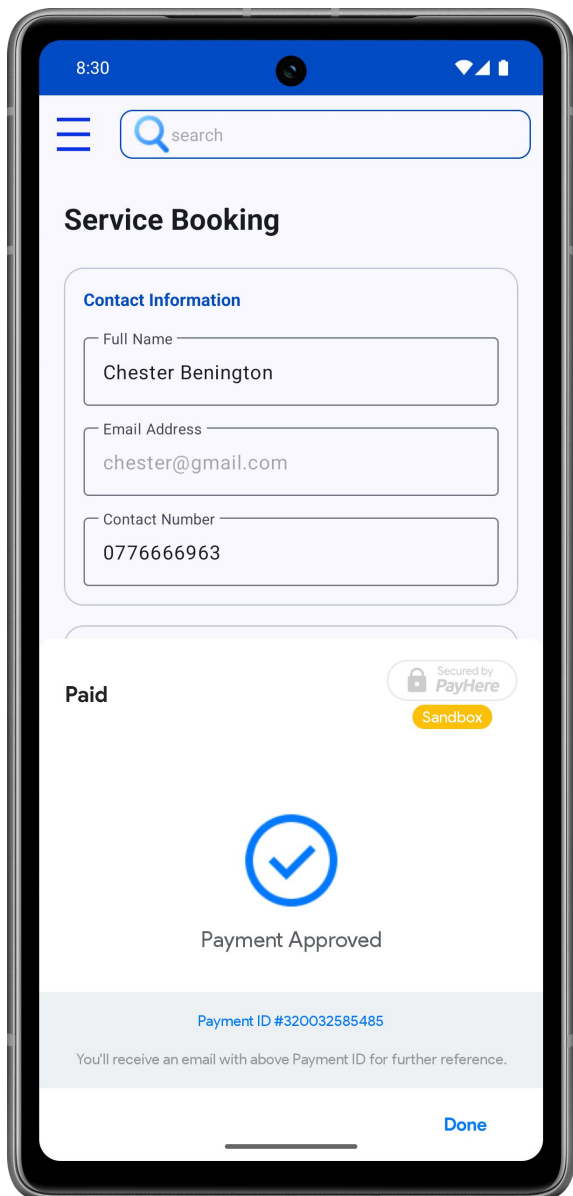
**Right Screen (8:26):**

- Service Booking** header.
- Bank Card** section, featuring a 'Secured by PayHere' badge and a 'Sandbox' label.
- Form fields for:
  - Name on Card
  - Credit Card Number
  - CVV (with a question mark icon)
  - Expiry MM/YY
- A large orange **Pay** button at the bottom.

**Figure 23: Payment Phase**

User allows to fill the card details and make the payment.

## Payment confirmation and storing booking



**Figure 24 Payment confirmation**



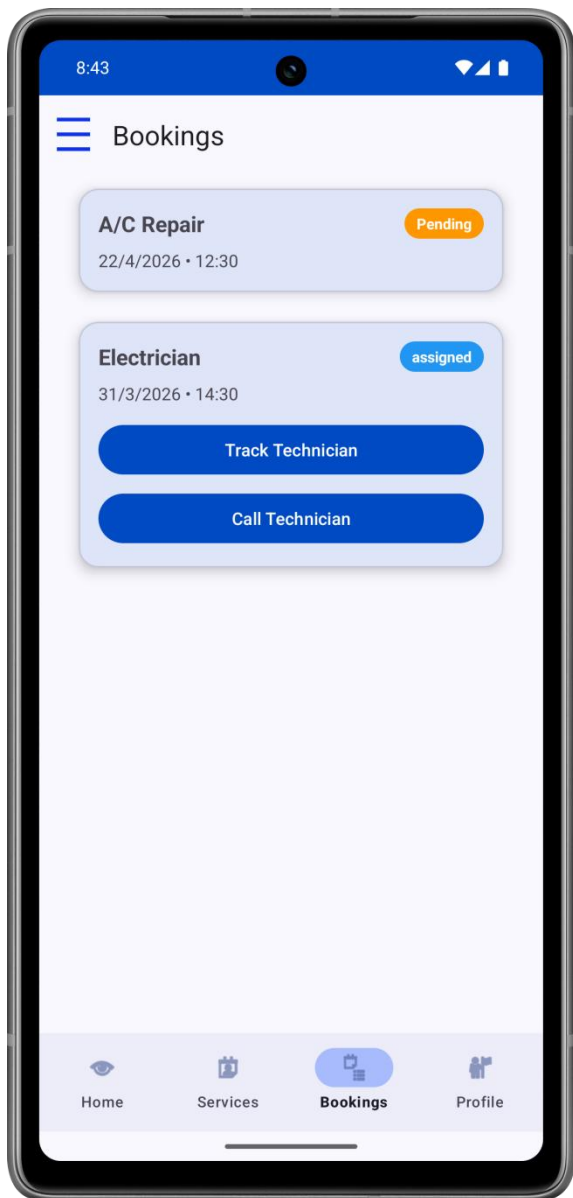
## Profile Screen

The image displays two mobile application screens for a 'Profile' section. Both screens have a blue header bar with the time '8:31' and status icons. The left screen shows the initial profile view, featuring a circular profile picture of a man with glasses and a beard. Below the picture, the title 'Profile' is followed by a section titled 'Personal Details' containing input fields for 'Full Name' (Chester Benington), 'Email Address' (chester@gmail.com), and 'Mobile Number' (0776666963). Below this is a 'Location Details' section with input fields for 'Address Line One' (No.1, Bandaranayaka Pura), 'Address Line Two' (Rajagiriya), and 'City' (Colombo). The right screen shows the edit profile view, where the profile picture is replaced by a large empty input field. The 'Email Address' field contains 'chester@gmail.com', and the 'Mobile Number' field contains '0776666963'. The 'Location Details' section is identical to the left screen. A prominent blue button labeled 'Save My Details' is located at the bottom of the form area. Both screens feature a bottom navigation bar with four icons: 'Home', 'Services', 'Bookings', and 'Profile' (which is highlighted in blue).

Figure 25: Profile Screen

In this screen user can view their personal details and user allows to update their details.

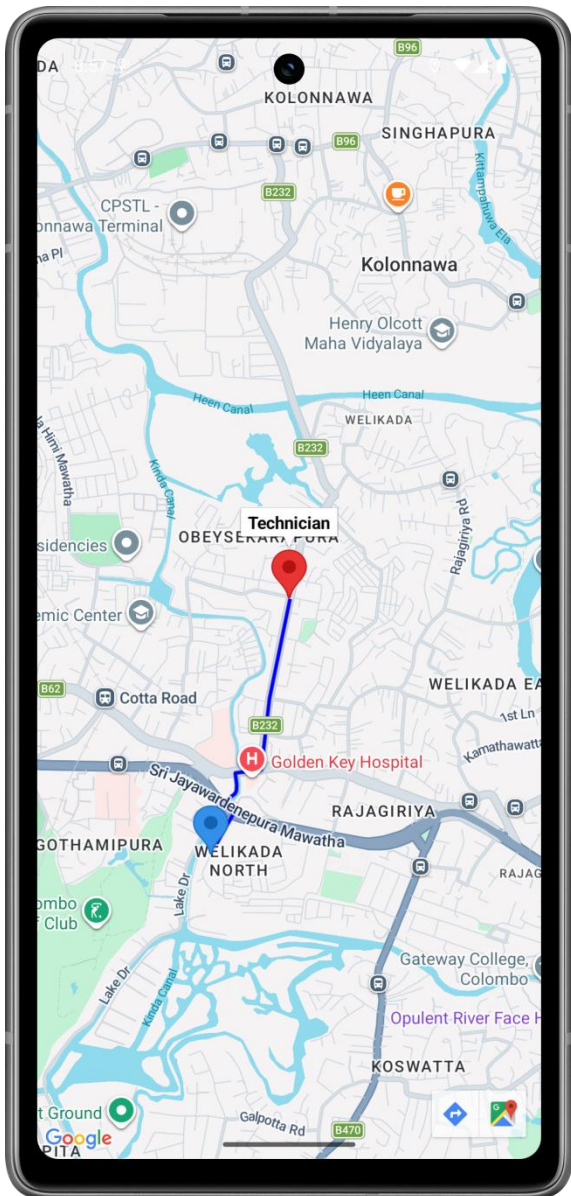
## Booking History



**Figure 26: Booking History**

Customer can see booking history and status. After technician assignment, customer allows to track the technician and call to technician. When customer shake the mobile, booking list will be refresh.

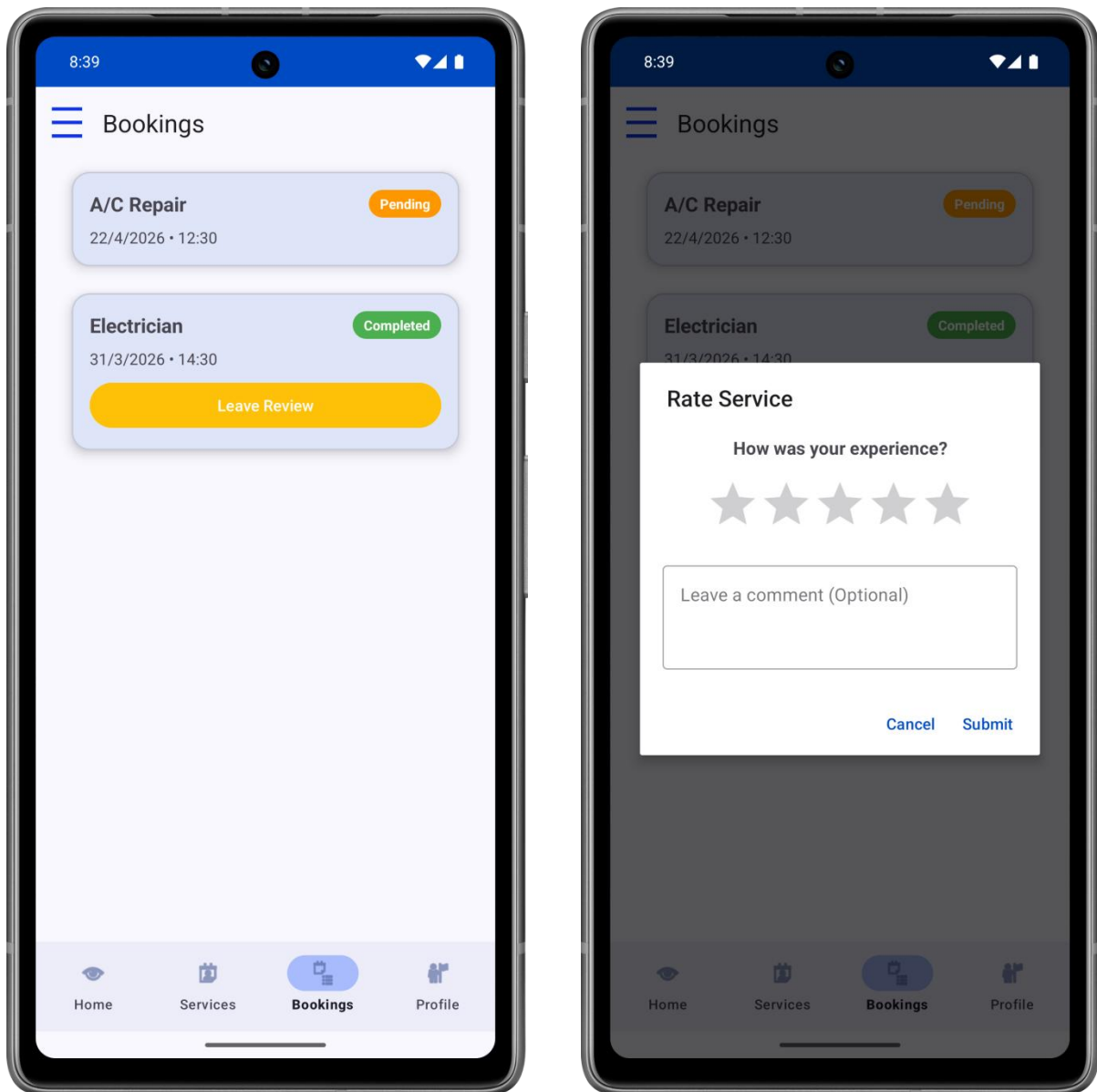
## Map



**Figure 27: Map**

Customer can track technician on Google map. The marker of technician is moving in real time when technician moves.

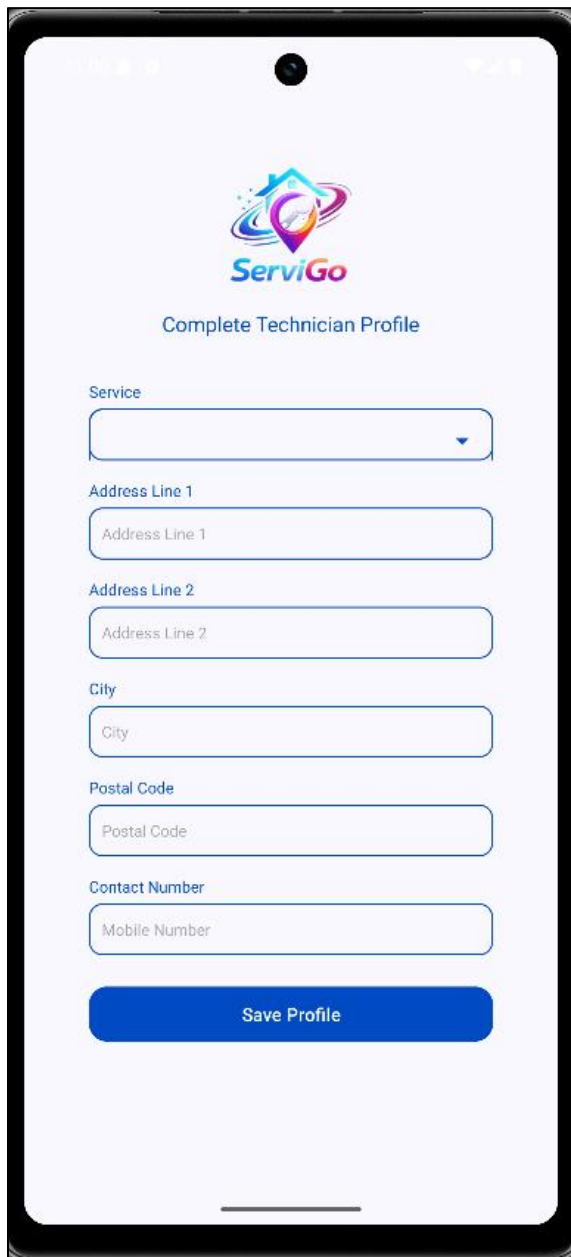
## Review Section



**Figure 28: Review Section**

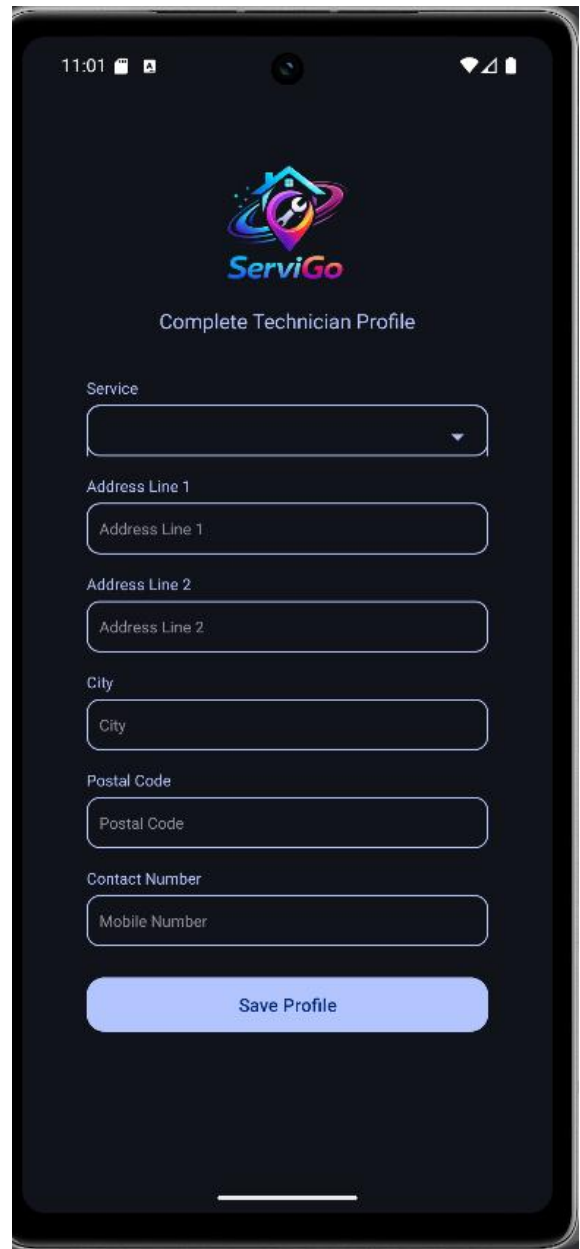
When the service is completed, customer can rate the service and leave comment.

## Complete Technician Profile Screen



The screenshot shows the 'Complete Technician Profile' screen in light theme. At the top is the ServiGo logo and the title 'Complete Technician Profile'. Below this are several input fields: a dropdown menu for 'Service', and text boxes for 'Address Line 1', 'Address Line 2', 'City', 'Postal Code', and 'Contact Number' (with a placeholder 'Mobile Number'). A blue 'Save Profile' button is at the bottom.

**Figure 29 : Complete Technician Profile Screen  
Light Theme**



The screenshot shows the 'Complete Technician Profile' screen in dark theme. It features the same layout as Figure 29, with the ServiGo logo, title, and input fields for 'Service', 'Address Line 1', 'Address Line 2', 'City', 'Postal Code', and 'Contact Number' (with a placeholder 'Mobile Number'). A light blue 'Save Profile' button is at the bottom.

**Figure 30 : Complete Technician Profile Screen  
Dark Theme.**

This is the Complete Technician Profile Screen of the ServiGo App. After the sign up as technician, user must create profile with details such as service, address, city, postal code and contact number.

## Technician Home

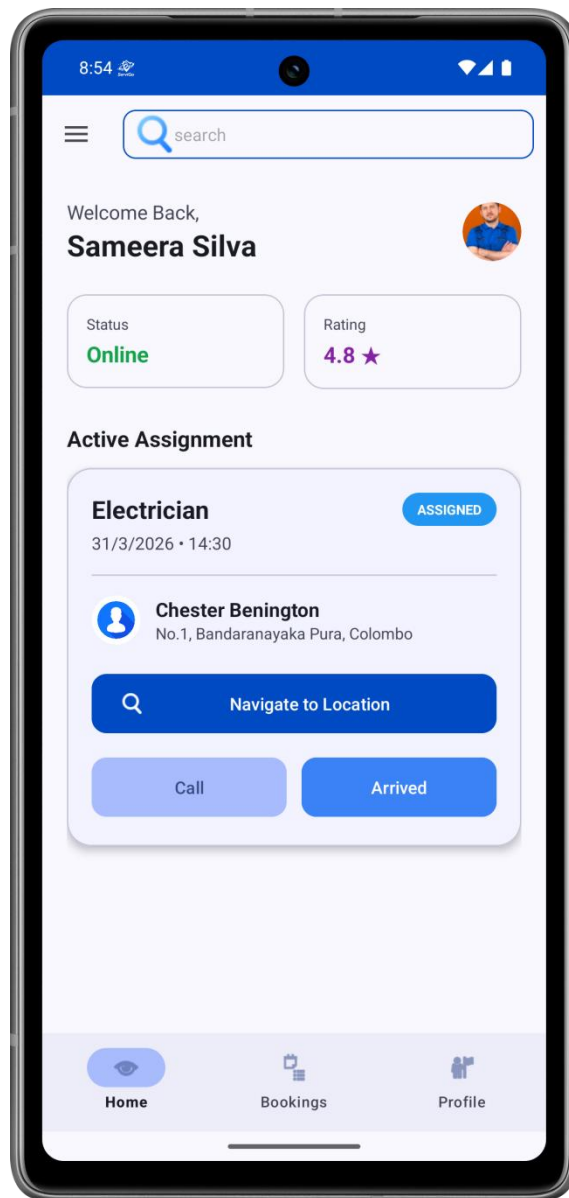
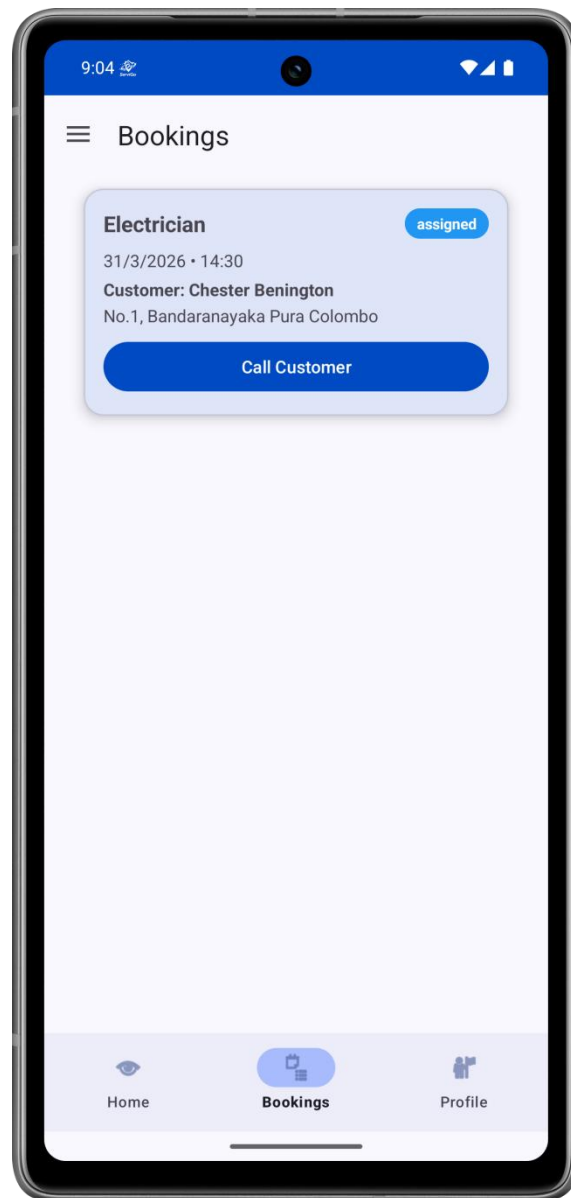


Figure 31: Technician Home

Technician can view assigned bookings from home screen, and can view the customer location on Google map. Technician allows call customer. After technician arrived, technician should press arrived button. Then booking status will be change.

## Booking History



**Figure 32: Technician booking history**

When technician shake the mobile, booking list will be refresh.

## Status Changing of booking

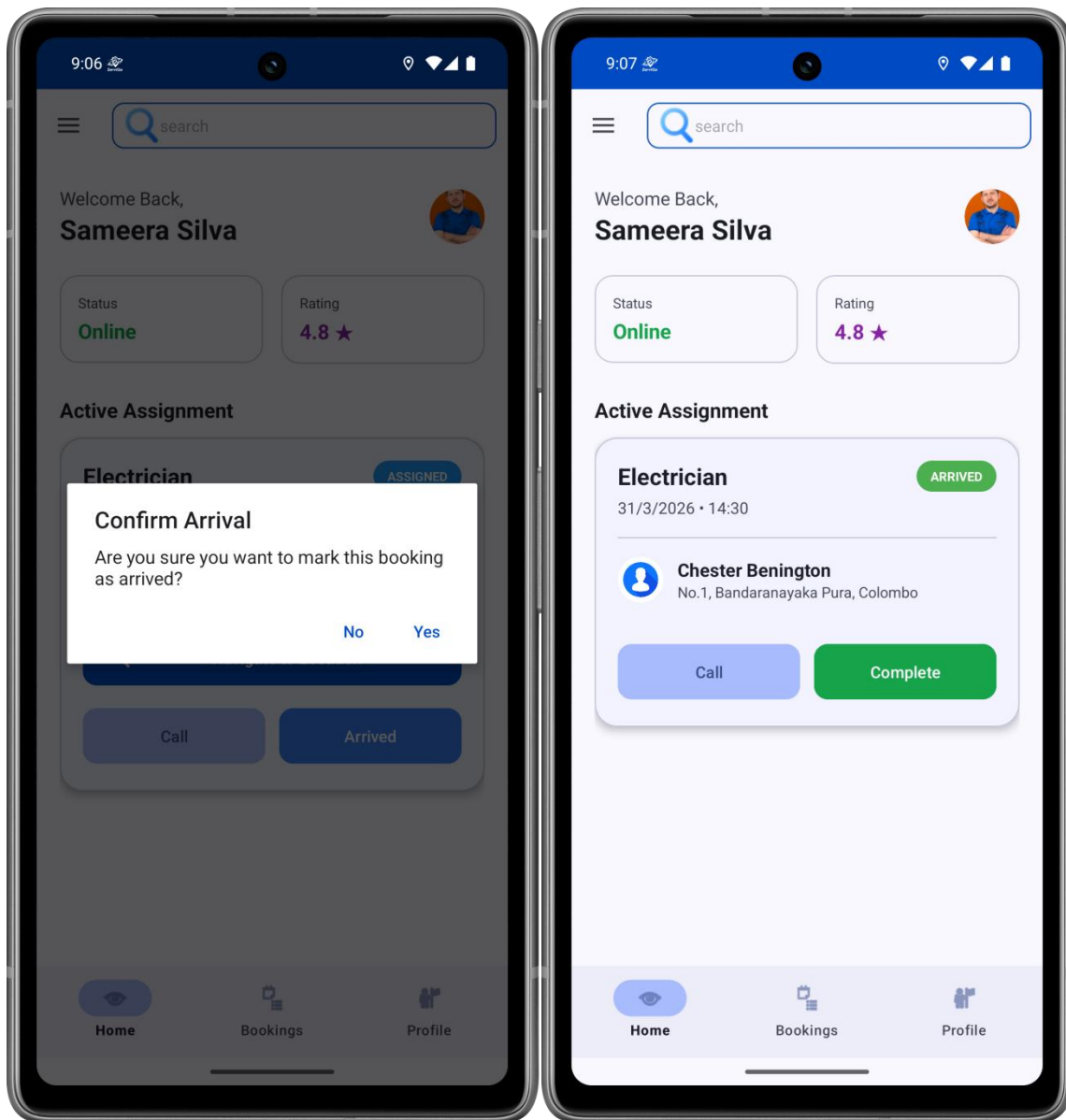
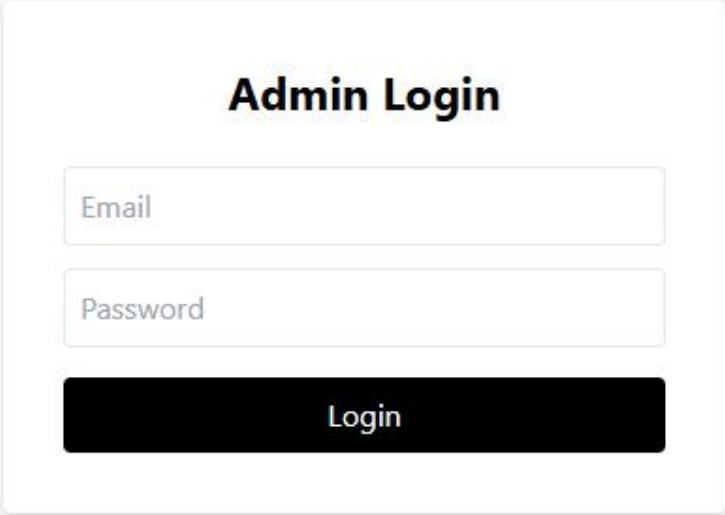


Figure 33: Status Changing

After technician arrived, technician should press arrived button. Then booking status will be change and show the complete button to press.



## Admin Login

The image shows a web form for admin login. It is centered on a light gray background. The form itself is a white rectangle with a thin gray border. At the top of the form, the text "Admin Login" is written in a bold, black, sans-serif font. Below this title, there are two input fields. The first field is labeled "Email" in a light gray font and has a white background with a thin gray border. The second field is labeled "Password" in a light gray font and also has a white background with a thin gray border. Below these two fields is a solid black rectangular button with the word "Login" written in white, centered text.

**Figure 34: Admin Login**

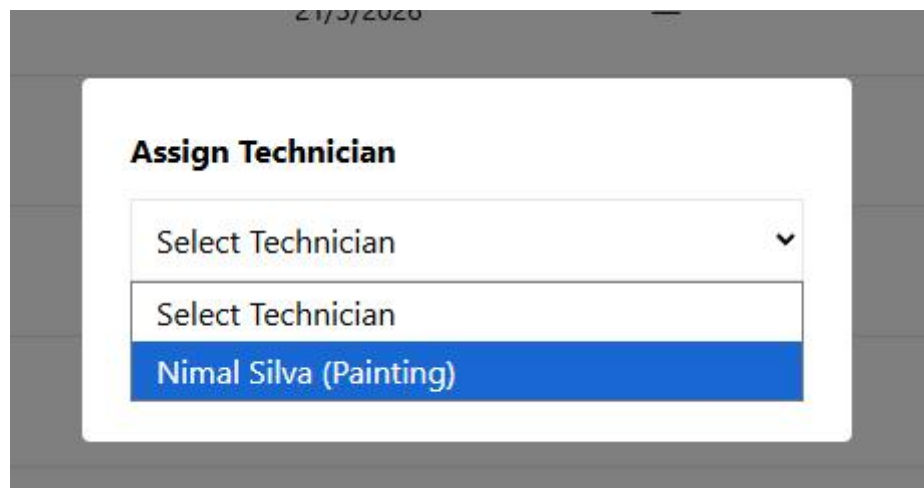
Admin can log in to the laravel based admin panel using this page.

## Booking Management

| ID | Service     | Customer                            | Date      | Technician    | Status    | Actions                                                              |
|----|-------------|-------------------------------------|-----------|---------------|-----------|----------------------------------------------------------------------|
| 2  | A/C Repair  | Devin Liyanapathirana<br>0743666996 | 21/3/2026 | —             | Pending   | <a href="#">Assign</a> <a href="#">Status</a> <a href="#">Delete</a> |
| 7  | Painting    | Tom Jay<br>0776999636               | 26/3/2026 | —             | Pending   | <a href="#">Assign</a> <a href="#">Status</a> <a href="#">Delete</a> |
| 8  | Painting    | Tom Jay<br>0776999636               | 31/3/2026 | —             | Pending   | <a href="#">Assign</a> <a href="#">Status</a> <a href="#">Delete</a> |
| 9  | Painting    | Tom Jay<br>0776999636               | 31/3/2026 | Nimal Silva   | Completed | <a href="#">Assign</a> <a href="#">Status</a> <a href="#">Delete</a> |
| 10 | Painting    | Tom Jay<br>0776999636               | 28/3/2026 | Nimal Silva   | assigned  | <a href="#">Assign</a> <a href="#">Status</a> <a href="#">Delete</a> |
| 14 | Electrician | Chester Benington<br>0776666963     | 31/3/2026 | Sameera Silva | arrived   | <a href="#">Assign</a> <a href="#">Status</a> <a href="#">Delete</a> |
| 15 | A/C Repair  | Chester Benington<br>0776666963     | 22/4/2026 | —             | Pending   | <a href="#">Assign</a> <a href="#">Status</a> <a href="#">Delete</a> |

**Figure 35: Booking management(Dashboard)**

Admin able to manage bookings such as, update status, technician assignment and delete bookings. In this page, admin can view count of all bookings, completed bookings and pending bookings. After technician assignment, customer get email of assignment.



**Figure 36: Technician Assignment**

This pop up loads filtered technicians based on booking service

## User Management

ServiGo

Dashboard

Users

Logout

User Management

Refresh List

Total Users  
6

Customers  
3

Technicians  
3

Search users by name, email...

| ID                            | Name                  | Email & Mobile                  | Role       | Status | Actions |
|-------------------------------|-----------------------|---------------------------------|------------|--------|---------|
| CucIm501VChbFASPhH0XINBoiPvF3 | Nimal Silva           | nimal@gmail.com<br>0778558585   | Technician | Banned | Unban   |
| Su77WoCa1gA7K3uO2UfQNKTCo2    | Chester Benington     | chester@gmail.com<br>0778666993 | Customer   | Active | Ban     |
| VhHcbrTA9eYsoAR45kz19J6Rk1    | Tom Jay               | tom@gmail.com<br>0776999636     | Customer   | Active | Ban     |
| dbH69TmM8hSbdd0DwU3JzSZVv2    | Prasanna Jayawardena  | prasanna@gmail.com              | Technician | Active | Ban     |
| g9UC7R9C3XP0s6Bz8nF2uOMBqn1   | Devin Liyanapathirana | devin@gmail.com                 | Customer   | Banned | Unban   |
| uVCj8dv6e0a7aXb5BRKqkDbkmu53  | Sameera Silva         | sameera@gmail.com<br>0740385256 | Technician | Active | Ban     |

Figure 37: User management

Admin can see total users, customer count and technician count. Admin allows to ban an unban users from this page.

## **5. Implementaion**

### **5.1. Backend Implementation**

To develop the backend of ServiGo uses PHP and follows the RESTful API. It helps to improve system central performance of ServiGo. This backend is handles all the requests from the client and responses, It communicates with mobile application, laravel based admin panel, MySQL database , Firebase Firestore and handles business logics.

This backend is used to handle some main operations of the ServiGo system such as store bookings, get booking details, updates status of bookings (in Technician side and admin side), technician assignments and delete bookings. Additionally the backend used to change users status like ban and unban. The ServiGo system shares data by using HTTP methods, like POST and GET. And it shares data in JSON format.

In additionally, the admin panel of ServiGo is developed with backend by using Laravel framework and it follows MVC architecture. It enable admins to assign technicians to each booking, update status of booking, delete booking, ban and unban users. In this phase represent this project is uses clean and well structured backend API and that ensures efficiency, scalability of the ServiGo system. And is is helps to communicate with other services.

### **5.2. Frontend Implementation**

The main front end is mobile application. The mobile application is designed by using java. The interfaces of mobile application is designed by using XML layouts with material design models. This combination is ensures the ServiGo provide modern, clean and smooth frontend.

The frontend of this project uses fragment based architecture with activity containers. With this architecture, user can navigate fast between other screens and it helps to manage better memory.

The mobile application that developed by using java is the frontend of the servigo project. The interfaces of serviGo is developed by using XML layouts with material design principles. This ensures that the interfaces gives are clean and user friendly experience. Frontend used RecyclerView for service listing, ViewPager2 for image slider, Customized input fields for user friendly interfaces. This ensures this ServiGo front end display dynamic data by using that components. Also View Binding is used to interaction with UI elements.

### **5.3. Database Design**

ServiGo uses both MySQL, Firebase Firestore and Firebase Storage to manage data. MySQL database is used to manage booking data, manage customer review data and admin data. Firebase Firestore is used to manage users data (technicians and customers data), service data and technician location data. Friebase storage is used to manage users profile pictures and image slider pictures. The URLs of users profile picture are storing in users collection in firebase firestore.

## 5.4. API Endpoints (PHP Backend)

| Endpoint                  | Method | Description                                                                           |
|---------------------------|--------|---------------------------------------------------------------------------------------|
| saveBooking.php           | POST   | Saving new bookings after customer confirm it with payment.                           |
| getBookings.php           | GET    | Fetch bookings for display it in customer's booking history.                          |
| getTechnicianBookings.php | GET    | Retrieve technician assigned bookings for display it in technician's booking history. |
| updateBookingStatus.php   | POST   | Update booking status by admin and technician.                                        |
| submitReview.php          | POST   | Save user feedback after submit a review.                                             |
| getReviews.php            | GET    | Retrieve reviews for a service to display it in single service details screen.        |

**Table 3: API Endpoints**

## 5.5. Security Measures

Several measures are used in ServiGo to protect users, booking and service data.

- ServiGo managing user sign in and sign up process with Firebase Authentication.
- ServiGo is developed using role based accessing. Only customer allows to see promotion banners, image sliders and popular services list. That ensures authorization.
- All the inputs of ServiGo is validated and it prevent the entering invalid and malicious data to the system.
- ServiGo sharing data between client and server through secure API.
- Admin allows to manage users like ban unban from the system securely.

## 5.6. Deployment

ServiGo is deployed with multiple platforms

- Android Application: Installed on user device as packaged APK
- Backend API: Hosted on web server with apache
- Database: MySQL server configured
- Firebase Services: Configured for authentication, storage, real time database, push notifications.

## **6. Result and Evaluation**

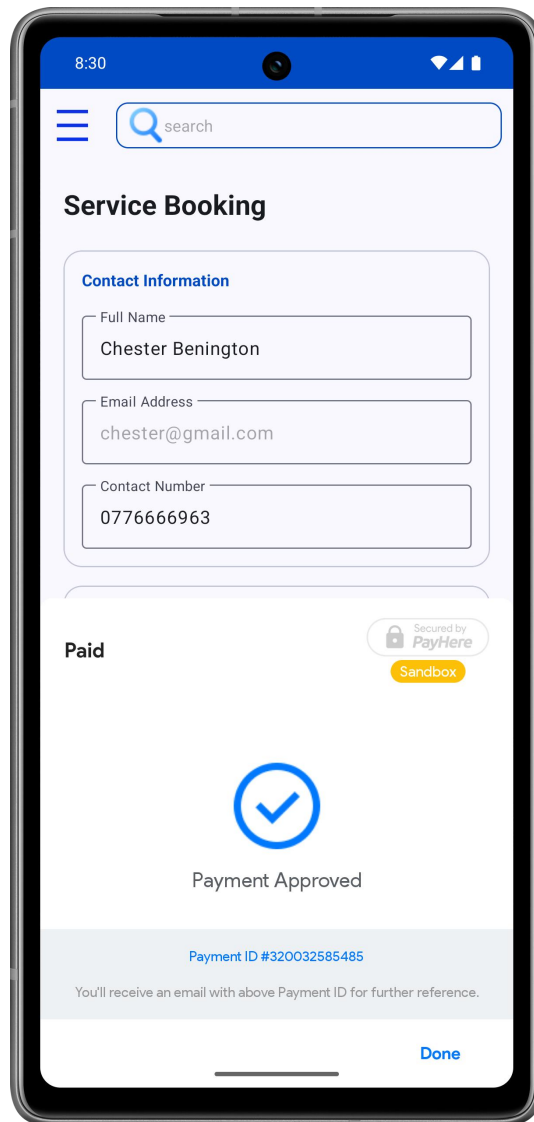
ServiGo system was developed successfully and tested to analyse system performance, functionality, efficiency and reliability. The goal of evaluation is to ensure the system meets the requirements that mentioned in earlier stage.

- Unit testing: This app used unit test to ensure individual components such as API endpoints, android modules are working as expected.
- Integration testing: Integration testing used to ensure good communication between API, mobile app and databases.
- System Testing: The complete work flow of the ServiGo is verified by using system testing.
- User Acceptance testing: The system is with real group of users, and that users verified the system meets business requirements and ready for real world production.

### **6.1. System Functionality testing**

The main feature of the ServiGo such as service booking, payment processing, real-time tracking, and admin management were tested. It ensures ServiGo system allows:

- Users to sign in and sign up successfully without any confusion and error by using Firebase Authentication.
- Customer to search services and schedule a booking after get an idea from other customers reviews without any errors.
- Admin to see that the bookings are listing successfully without error by retrieving data from the backend database.
- Customer to make payment to confirm booking through the payhere without physical cash securely. The bookings are stored only after the confirmation payment.



**Figure 38: Payment approved**

- Customer to track technician in real time through Google map accurately.
- Admin functionalities such as assigning technicians, updating booking status, and managing users worked as expected.

## 6.2. System performance

API response and requests are performing successfully as expected. And the firebase real time data updates ensure minimal delay. Push Notifications are also performed as expected.

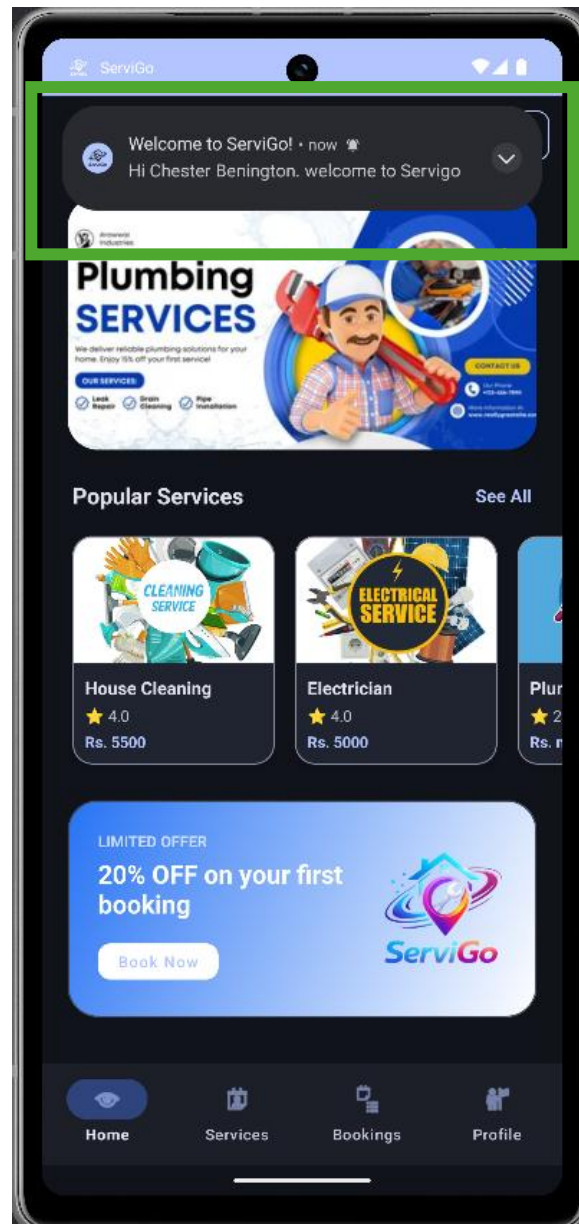


Figure 39: Push Notification

## 6.3. User Experience Evaluation

The ServiGo system is tested by a group of users.

- User found the simple user friendly interface after login and user navigate between the screens easily without any confusion.
- Some users search services on search bar, some users navigate to services screen and viewed the service details in single service details screen.



- Users got a ideas about services from ratings and reviews before schedule a booking.
- Real time updates on firebase is help to improve users trust, confidence and transparency of the service.

#### **6.4. Limitaions**

- The system mainly depends on the internet. Because for the system is runs based on real time data.
- ServiGo only provide single payment gateway.
- Process of technician assignment is done manually by the administrators. It is not good for large scale usage.

#### **6.5. Overall Evaluation**

The ServiGo system is meets the requirement and outputs as expected. This system uses modern technologies such as firebase real time updates, mobile payments, real time tracking. That ensures this system is provide complete solution for service booking, real time tracking and simplify the users tasks.

### **7. Future Work**

- Add more payment gateway for more flexibility
- Implement multi language support
- Current application allows get calls between technicians and customers. But there is no chatting channel. Crate chatting feature with firebase will be more useful in the future.
- Implement AI, for get AI based recommendations
- Add a dashboard to view earnings of technicians.
- Advanced reporting features for administrator
- Automatically technician assignment

### **8. Conclusion**

ServiGo project started with two goals. To show Sri Lanka home service market is ready for a new step with ServiGo mobile application. And This project prove a one student of a university could develop the system within a university project.

Both goals are turn out to be true but it was not easy. The market gap is clearly real because there is no platform for home service marketplace in Sri Lanka. That is a problem worth solving. ServiGo fills that both technical gap and practical gap. On technical side, ServiGo is fully build and tested. The app uses firebase , Google map, online payments, live location

tracking and push notification, motion sensors. Also uses web site for the admin panel. These works together smoothly as one experience.

The main highlight of the ServiGo is live technician tracking screen. Users able to see the technician marker and route on map and moving it with technician in real time. Additionally uses scalable technology such as RESTful APIs that ensures system can manage future improvements.

## **9. Reflection**

The development of the ServiGo project is more valuable learning experience. It improved my problem solving skills and the technical skills. Through this project I got experience in developing mobile app using backend services and web based admin panel.

In this project I was understanding how to work with different technologies for a one project. I was learned to connect android mobile app with Firebase authentication, PHP base REST API and implement a Laravel website for admin panel. This helps to understand the architecture of full stack development.

Additionally improved my skill to design user friendly, simple and responsive interfaces. Development of real time tracking and online payments are lead me to improve my logical thinking and manage work flow.

I also learned how to debug errors, manage performance, make reliable and efficient application. This project improve my skill to adapt to the challenges and analytical thinking.

## 10. References

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